Daniel Raymond

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Education

Aug 2022 – May 2026 B.S. (Electrical & Computer Engineering) at **Worcester Polytechnic Institute** (GPA: 3.50/4.0)

Publications

Fuller, Brianna et al. (Oct. 2024). *Developing Signage for the Cowasuck Band of the Pennacook-Abenaki People's History and Stories*. Tech. rep. 100 Institute Road, Worcester MA 01609-2280 USA: Worcester Polytechnic Institute.

Broomberg, Leslie et al. (Jan. 16, 2026). "Devices and Methods for Managing Power of Modular Gas to Liquid Systems". patentapp 19/452,116 (United States). Emvolon Inc. Pending.

Work Experience

Burlington Electric Department *Burlington, VT* *Generation Generalist II* **June 2026 - Present**

- Unloaded biomass from train cars in various conditions, ensuring safety and communication.
- Assisted mechanics and electricians in troubleshooting, repairs, and maintenance of conveyors, electrostatic precipitators, and water chemestry systems.
- Followed preventative maintenance processes and completed operator trainings on grid-level generation equipment.

Emvolon Inc. *Woburn, MA* *Electrical Engineer Intern* **May 2025 - Aug 2025**

- Deployed **IEC 61131-3** PLCs with intrinsically safe sensors and actuators to control hazardous materials
- Programmed a FOC synchronous **motor drive** in **C** to balance energy flow between an engine and **battery bank**
- Created line diagrams with **SolidWorks Electrical** to implement system requirements derived from **SOPs** and P&IDs

Jay's Garage *Mendon, MA* *Automotive Technician* **Jun 2023 - Aug 2023**

- Diagnosed fickle **CAN bus** faults and quiescent current draws using **schematics**, voltmeters, and **oscilloscopes**
- Performed **component-level repairs** using reverse engineering techniques that saved on ordering expenses and lead times

Studio David Teeple *Northampton, MA* *Studio Assistant* **Jul 2022 - Present**

- Helped relocated a sculpture studio including 30 tons of materials, requiring micro- and macro-level **planning**
- Collaborated to **configure workspaces** that allowed a densely filled studio to function creatively and practically

Hennessey Precision *Northampton, MA* *Business Assistant* **Aug 2020 - Feb 2022**

- Assisted the owner in validated issues, ordering parts, completing repairs, and **billing clients** to offload workload
- Increased shop throughput by 50% by utilizing two bays, enabling business growth into a larger space

Projects

Permalux LED Lamp **Mar 2025 - May 2025**

- Produced an LED desk lamp incorporating a user-replaceable CoB driver to avoid product obsolescence
- Analyzed the manufacturing cost, **failure rates**, and market value which proved profitability at a $100 retail price
- Built a **switch-mode PSU** with high power factor and low THD to reduce component stresses and prevent flicker

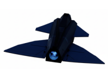
USB PD Charger **Apr 2024 - Present**

- Building a first of a kind charger powered by generic DC sources that supports the full USB PD profile up to 240W
- Programming **mixed-signal control loops** with stability targets to ensure adherence to protocol specifications
- Testing of a prototype showed efficiency of 90%, attributable to **GaN FETs** and **multilayer PCB design**

Discrete Component Opamp **Mar 2024 - May 2024**

- Simulated a **rail-to-rail opamp** with **analog design** principles and consideration for real-world constraints
- Implemented with individual **transistors** and tested against **SPICE simulation** to validated performance

Hybrid Electric RC Jet **Dec 2019 - Aug 2021**



- Designed and **hand-fabricated** an RC airframe with integrated afterburner, extending practical flight time far beyond the sub-minute elecric limitation through a 500g custom propane system that boosted thrust by 1600g
- Applied **static analysis**, **hand calculations**, and **testbenches** to ensure **redundancy** and high FoS

Buck Voltage Regulator **Mar 2019 - Jun 2024**

- Innovated **digital logic** for a **multiphase regulator** that optimized transient response and quiescent power
- Iterated through 6 **hardware revisions** to achieve >98% efficiency and 1kW output power with high reliability

Extracurriculars

a-cappella, conductor, director, arranger | skiing, rock climbing, hiking, running | FPV racing, car modification, cooking